

MODESTO CITY, CA, USA

WMO: 724926

Lat: 37.624N

Lon: 120.951W

Elev: 22

StdP: 101.06

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 23258

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.4	0.9	-6.1	2.3	13.1	-4.0	2.7	12.4	10.1	12.1	8.7	11.3	1.0	120	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	38.7	20.8	37.0	20.2	35.5	19.5	22.0	35.7	21.1	34.6	20.4	33.5	4.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.9	12.1	28.1	15.8	11.3	26.5	14.9	10.6	25.4	64.2	35.8	61.0	34.7	58.4	33.6	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.6	6.9	DB	-2.5	41.5	1.2	1.4	-3.4	42.4	-4.1	43.2	-4.8	44.0	-5.7	45.0	(4)
(5)			WB	-3.0	23.2	1.3	1.1	-3.9	24.0	-4.6	24.7	-5.4	25.3	-6.3	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	8.7	11.0	13.8	16.3	20.3	24.1	26.5	25.6	23.4	18.3	12.5	8.7	(6)
(7)		DBStd	6.95	2.61	2.56	3.02	3.49	3.62	3.59	3.06	2.60	3.05	2.90	2.96	2.76	(7)
(8)		HDD10.0	145	56	17	4	1	0	0	0	0	0	0	9	58	(8)
(9)		HDD18.3	1256	297	204	142	80	20	3	0	0	2	36	174	298	(9)
(10)		CDD10.0	2870	16	46	122	189	319	422	510	484	401	257	86	19	(10)
(11)		CDD18.3	939	0	0	2	18	81	174	252	226	153	34	1	0	(11)
(12)		CDH23.3	12101	0	1	63	302	1072	2279	3322	2801	1799	448	15	0	(12)
(13)		CDH26.7	6378	0	0	7	97	476	1242	1953	1566	904	133	0	0	(13)
(14)	Wind	WSAvg	2.9	2.0	2.5	2.9	3.4	3.8	3.9	3.7	3.4	2.9	2.5	1.9	2.2	(14)
(15)	Precipitation	PrecAvg	314	56	47	53	26	4	2	1	2	7	19	51	43	(15)
(16)		PrecMax	696	144	149	165	75	28	12	18	19	56	62	153	100	(16)
(17)		PrecMin	162	5	1	0	3	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	122	40	40	42	22	6	3	4	5	15	17	38	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	22.7	27.7	32.8	36.8	40.2	41.0	39.7	38.0	33.1	25.2	18.3	(19)
(20)			MCWB	11.9	14.4	16.0	18.1	19.8	21.4	21.8	20.7	19.7	17.8	15.8	12.9	(20)
(21)		2%	DB	17.0	20.6	25.0	29.3	34.2	37.8	38.9	37.6	35.9	29.9	23.0	16.4	(21)
(22)			MCWB	11.9	13.6	15.4	16.5	18.6	20.3	21.0	20.3	19.2	17.0	14.6	11.8	(22)
(23)		5%	DB	15.3	18.5	23.0	26.9	31.8	35.5	37.1	35.8	34.0	27.9	20.6	15.1	(23)
(24)			MCWB	11.2	12.5	14.5	15.5	17.4	19.4	20.4	19.8	18.8	16.0	13.7	11.4	(24)
(25)		10%	DB	13.7	16.7	20.8	24.3	29.1	33.3	35.3	34.1	32.0	25.9	18.5	13.8	(25)
(26)			MCWB	10.5	11.8	13.5	14.6	16.4	18.6	19.7	19.4	18.2	15.4	12.7	10.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.7	17.7	19.1	21.2	22.8	23.7	22.6	22.3	19.3	17.2	15.1	(27)
(28)			MCD B	16.7	19.9	25.0	30.0	34.7	38.0	38.1	36.7	33.1	29.2	23.2	17.0	(28)
(29)		2%	WB	12.9	14.5	16.3	17.5	19.5	21.3	22.2	21.4	20.5	18.1	15.7	13.5	(29)
(30)			MCD B	15.0	18.9	23.3	27.7	32.3	35.3	36.3	34.9	32.4	27.5	20.9	15.2	(30)
(31)		5%	WB	11.9	13.4	15.2	16.3	18.2	20.2	21.2	20.6	19.6	17.1	14.6	12.3	(31)
(32)			MCD B	14.1	17.4	21.6	25.1	30.0	33.5	35.0	33.8	31.7	25.6	19.0	14.0	(32)
(33)		10%	WB	10.9	12.4	14.1	15.1	17.0	19.3	20.4	19.8	18.8	16.2	13.6	11.1	(33)
(34)			MCD B	13.1	16.0	19.7	22.8	27.6	31.9	33.8	32.7	30.7	24.1	17.4	13.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	10.6	11.9	13.1	14.8	16.2	16.8	16.5	15.8	14.2	11.5	9.3	(35)
(36)			MCDBR	12.2	13.7	15.4	17.4	18.1	18.4	17.8	18.0	18.3	17.6	14.7	11.1	(36)
(37)			MCWBR	7.4	7.7	7.1	7.0	6.5	6.3	6.3	6.3	6.0	6.7	7.4	6.8	(37)
(38)		5% WB	MCD BR	8.8	12.0	14.1	15.9	17.2	17.5	16.9	16.9	16.8	15.3	12.3	7.1	(38)
(39)			MCWBR	5.8	7.2	7.0	7.0	6.7	6.6	6.6	6.3	6.0	6.2	6.6	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.335	0.341	0.358	0.362	0.355	0.366	0.350	0.352	0.353	0.343	0.326	(40)	
(41)		taud	2.434	2.465	2.440	2.393	2.400	2.430	2.412	2.437	2.436	2.436	2.435	2.492	(41)	
(42)		Ebn at Noon	839	896	923	921	918	920	908	918	898	861	826	831	(42)	
(43)		Edh at Noon	86	94	105	117	118	115	116	110	104	95	85	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.32	3.36	4.80	6.21	7.46	8.34	8.18	7.39	6.05	4.36	2.87	2.10	(44)	
(45)		RadStd	0.37	0.40	0.46	0.53	0.41	0.29	0.23	0.20	0.19	0.26	0.18	0.32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-142	+177	N/A	(46)
(47)	Regional (47 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-120	+138	+32	(47)

Nomenclature: See separate page